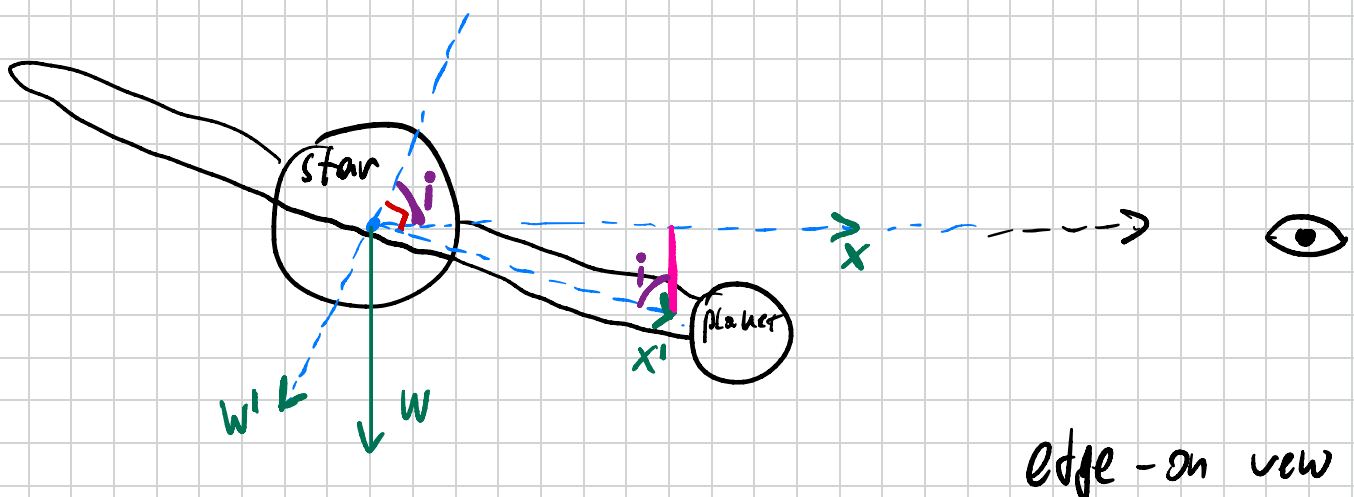
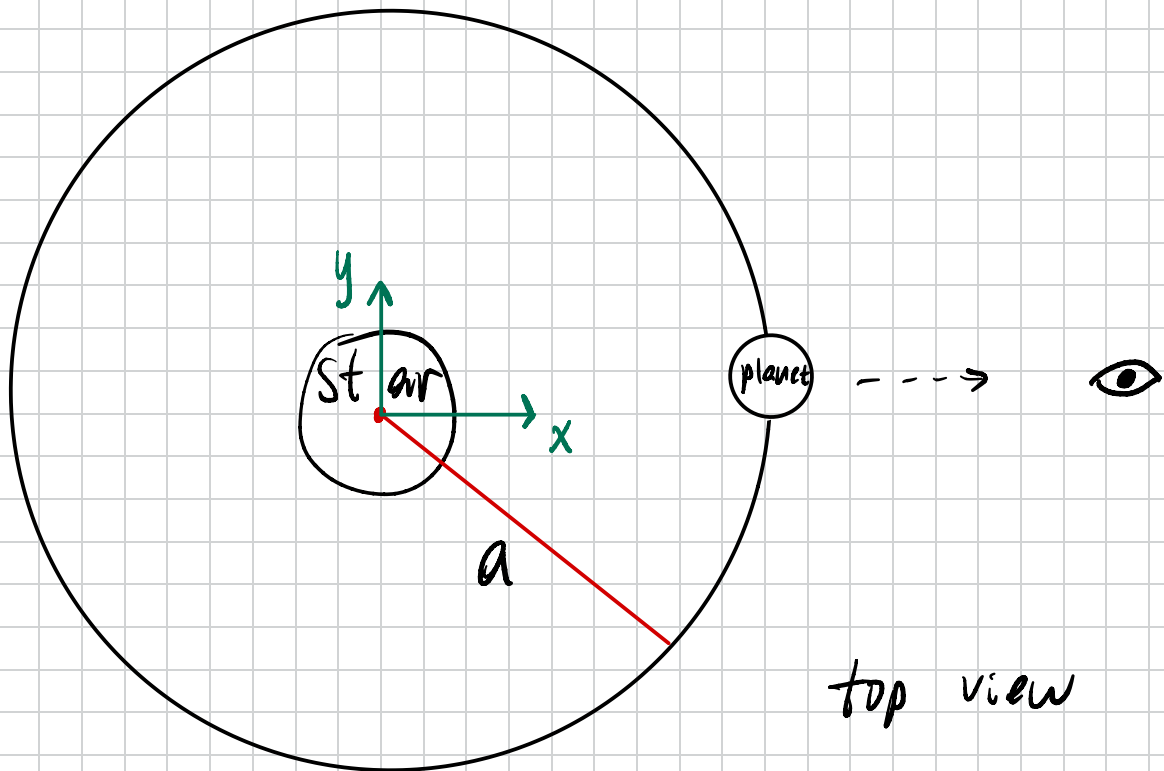


## 4. Orbital Separation

Suppose  $\omega$  is the orbital frequency (T-period),  
 $a$  is the orbital radius, and  
 $t$  is time measured from mid-transit,  
what is  $z(t)$ . We need it to plot the  
light curve (relative flux  $F(p, z)$  against time)



I will operate with real distance  $d = z r_*$  for derivation. Then will divide by  $r_*$  for  $z$ .

First, let us consider simplest case:  $i = 90^\circ$   
Assume the planet in the 1st drawing moves up, so if we plot its coordinates on the  $x$ - $y$  plane, we get

$$x(t) = a \cdot \cos\left(\frac{2\pi}{T} t\right) = a \cdot \cos(\omega t)$$

$$y(t) = a \cdot \sin\left(\frac{2\pi}{T} t\right) = a \cdot \sin(\omega t)$$

$$d = y(t) = a |\sin(\omega t)|$$

Now, we add the tilt (rotation about  $y$ -axis), which introduces the third coordinate:  $w$ .

So in the  $(x', y, w')$  system, the coordinates are  $(a \cdot \cos(\omega t), a \cdot \sin(\omega t), 0)$ .

We need to convert it to  $(x, y, w)$ .

Then, since observer is sitting on the  $x$ -axis, this coordinate is irrelevant  $\Rightarrow$

$\Rightarrow$  answer will be  $\sqrt{y^2 + w^2}$

$x' \rightarrow x$  converts  $a \cdot \cos(\omega t) \rightarrow a \cdot \cos(\omega t) \cdot \sin(i)$

$w' \rightarrow w$  converts  $0 \rightarrow a \cdot \cos(\omega t) \cdot \cos(i)$

Therefore

$$d(t) = \sqrt{a^2 \sin^2(\omega t) + a^2 \cos^2(\omega t) \cos^2(i)}$$

$$\Rightarrow z(t) = \frac{a}{r_*} \sqrt{(\sin(\omega t))^2 + (\cos(\omega t) \cos(i))^2}$$

Note: this function will give exact values (including for transit) twice a period. As we don't want to record a dip in flux when planet is \*behind\* the star, apply a constraint:  $\cos(\omega t) > 0$ .